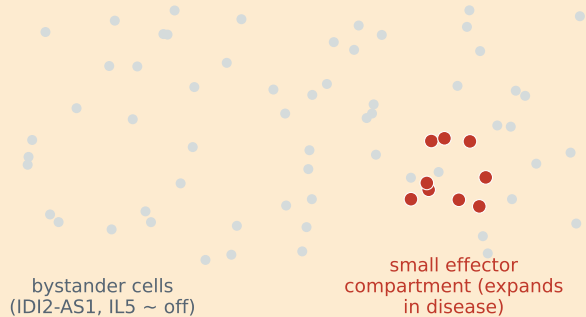


A cell-composition-adjusted re-analysis pipeline separates per-cell regulation from cell-composition shift in bulk RNA-seq

Worked example: the lncRNA IDI2-AS1 and its in vitro target IL5 across five public cohorts of type-2 allergic disease (n = 856)

THE PROBLEM

Bulk tissue mixes two signals



Per-cell IDI2-AS1 may fall, but the
effector compartment expands -> the
bulk average is masked or sign-inverted

THE PIPELINE

bulk-only, open source

1 Differential expression
+ random-effects meta-analysis

2 Sample-level IDI2-AS1 vs IL5
correlation

3 Cell-composition adjustment
(marker signature + xCell)

4 Effect decomposition
+ confounding sensitivity

THE RESULT (asthma, n = 695)

~70-90% of the bulk IDI2-AS1-IL5
covariation attributable to eosinophils



marker signature ~90% | xCell ~70% (both $p \leq 0.006$)
within-tissue residual component ~ 0

**=> A bulk null is compositional MASKING
of a per-cell hypothesis, not its refutation**

Next step: single-cell / spatial test
(the pipeline nominates; it does not validate)